ODE-FFN: Progress Report 1

Independent Study — Deep Learning

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1 Project and main progress

This project asks whether a dynamical feed-forward block better supports flow matching, which learns a velocity field transporting noise to data. The intended application is the FFN in JiT image generators; we first study the original ODELayer on interpretable unconditional synthetic problems.

Evidence now includes 30 training trajectories, 54 checkpoint diagnostics, earlier spectral-cap controls, **66 frozen-checkpoint evaluations**, and **48 solver-resolution checks**. A better exponential approximation is not necessarily a better learned block: exact substitution worsens generation in all six low-order checkpoint–target pairs. K8 is nearly exponential, while similar separately trained K4/K8 scores conceal functional differences. Spectral caps lower measured sensitivity without consistently improving samples.

2 What the original layer computes

For hidden input h, the layer computes one scalar duration and reuses P:

$$F_K(h) = T_K(s(h)P)h, \quad T_K(A) = \sum_{k=0}^{K} \frac{A^k}{k!}, \quad s(h) = 0.2 + 0.8\ \mathrm{sigmoid}(g(h)/10). \tag{1}$$

Here g is a two-layer SiLU network. ODEFFN applies two such layers with SiLU between them. At $K = 1$, $F_1(h) = h + s(h)Ph$. Holding s fixed, $\exp(sP)h$ solves $dy/du = Py$ for duration s. This hidden duration is distinct from the external flow-matching time t.

The mathematical distinction. The exponential series converges for every finite P; the issue is truncation and sensitivity. For $r = s\|P\|_2$, $\|\exp(sP) - T_K(sP)\|_2 \le e^r r^{K+1}/(K+1)!$. However, differentiating the actual input-gated layer yields

$$J_{F_K} = T_K(sP) + [P\,T_{K-1}(sP)h]\,\nabla_h s^\top. \tag{2}$$

Controlling P controls only part of the layer's sensitivity. A small remainder or hidden Jacobian need not minimize the generative error.

3 Experimental setting

Data. The targets are a 2D Gaussian with mean $(1, -1)$ and covariance $\mathrm{diag}(0.09, 4)$; eight equally weighted Gaussian modes on a radius-3 ring with within-mode covariance $0.04I$; and centered two moons with noise standard deviation 0.08. These are specified toy distributions, not an exact paper-benchmark reproduction. Each has 100,000/10,000/10,000 train/validation/test points. Independent source noise is standard Gaussian; t is uniform. The target velocity is $x_1 - x_0$ at $x_t = (1-t)x_0 + tx_1$, with no class labels.

Controls and metrics. All training uses seed 0, width 64, three residual blocks, batch 512, and AdamW at 3×10^{-4}; the default endpoint is 5,000 updates. Primary evaluation is SW_2 (lower is better): the root mean squared difference of sorted projections on 128 fixed directions, using 2,048 fixed generated/target points and 32 Heun steps. Gaussian and Ring8 also have analytical velocity references. These are single-seed observations, not significance tests.

4 E1: compare against standard FFNs at controlled budgets

Purpose. Control for computation. Parameter counts are close: ODE 82,056, SiLU 81,858 and SwiGLU 81,792. ODE $K = 2$ uses 105,152 forward linear MACs per example, versus 80,000 for SwiGLU. At 6,572 updates, SwiGLU approximately matches the cumulative forward-MAC budget of ODE $K = 2$ at 5,000 updates.

Observation (Table 1). ODE $K = 2$ beats same-update SwiGLU on all targets, but SiLU is better on Ring8. After the MAC match, ODE remains better only on Moons. We retain fixed endpoints: Ring8 SiLU changes from 0.1631 to 0.2258 between updates 5,000 and 5,024, illustrating endpoint sensitivity.

5 E2: separate numerical accuracy from learned adaptation

Purpose and intervention. Freeze each input-gated $K = 1, 2, 4, 8$ checkpoint and evaluate every ODELayer at orders $1, 2, 4, 8$ or with a direct matrix exponential. This gives 60 cells plus six unchanged controls. Operator actions use float64; the rest remains float32. All 18 native reproductions and same-order precision controls pass (maximum native/reference SW_2 drift 4.47×10^{-8}). Parameters are frozen, but downstream hidden states and gates can change. Local tests separately hold native inputs and gates fixed (Fig. 4).

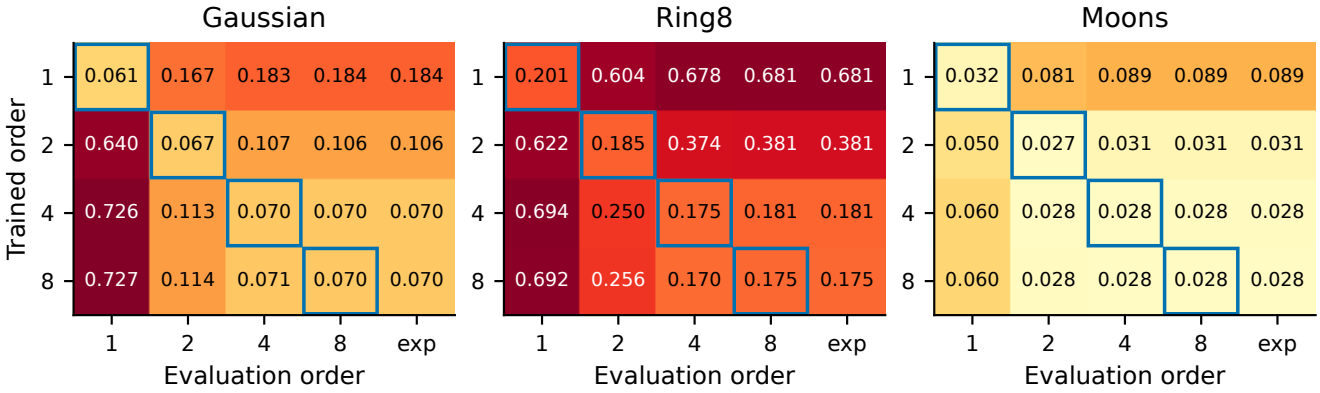


Figure 1: Full frozen-checkpoint SW_2 matrix. Rows: trained order; columns: evaluation order. Blue outlines mark the trained order. Shared logarithmic color scale; printed numbers are rounded. All use seed 0 and 5,000 updates.

Observation. Exact substitution worsens SW_2 in all six K1/K2 cases, by 16.1%–238.9%. For Ring8 K2, SW_2 rises from 0.18482 to 0.38133; oracle velocity MSE rises from 0.02871 to 0.16944. The latter change occurs before integration; endpoint mode structure also changes (Fig. 2). The checkpoint is adapted to its polynomial. This does not imply that a model *trained* with exponentials would perform poorly.

K4/K8 case study. Separately trained Ring8 K4/K8 scores differ by only 0.198%. Yet frozen K4-to-K8 worsens SW_2 by 3.225%, and K8-to-K4 improves it by 2.672%. K8-to-exponential changes SW_2 by at most 0.0030% across the three targets. K8 saturates here, but K4/K8 are not interchangeable. Training K8 costs 63.7% more forward linear MACs than K4.

Solver check. A post hoc Ring8 check repeats twelve modes at 32, 64, 128 and 256 Heun steps (two velocity evaluations per step). At 256 steps, K2-to-exponential still changes SW_2 from 0.18613 to 0.38371, and K4-to-K8 still worsens it by 3.1%. The intervention directions persist (Fig. 3); this is finite-resolution evidence, not a proof of numerical convergence.

6 E3: what should be constrained in P?

Spectral cap: purpose and observation. Earlier experiments compare free P with $P_{\text{eff}} = P \min(1, 1/\|P\|_2)$ at K1/K2. The cap lowers the measured full-layer Jacobian norm in all six comparisons but improves SW_2 in only three (Fig. 6). Ring8 K2's Jacobian norm falls from 3.014 to 2.099, while SW_2 worsens from 0.1848 to 0.1889. These norms average 16 observed inputs per layer and six layers; they are not global bounds. Smaller internal gain alone is an insufficient design objective.

Gate control. A time-conditioned gate removes the second term of Eq. (2) at fixed external time, while preserving gate-network size. After retraining, this does not uniformly reduce sensitivity or improve samples. For Moons K2, the Jacobian norm rises from 2.334 to 2.509, while SW_2 improves slightly from 0.02708 to 0.02663. This is not a pure derivative intervention: gate inputs and optimization change, and learned operators can compensate. It argues for measuring the whole block instead of inferring its behavior from one removed term.

Training trajectories. Across targets and K1/K2/K4/K8, the six-layer mean largest singular value grows from about 1.109 to 1.715–1.862, while stable rank $\|P\|_F^2/\|P\|_2^2$ falls from 17.37 to 8.31–9.40. Figure 5 shows K2. This means energy concentrates relative to the strongest direction, not that algebraic rank is lost. Flattening the spectrum could remove learned direction selection; its benefit must be tested rather than assumed.

Controlled matrix example: purpose and observation. To separate scale from geometry, construct $P_\theta = -\frac{1}{4}I + a_\theta N_\theta$ in three dimensions, with $(N_\theta)_{12} = \cos\theta$, $(N_\theta)_{23} = \sin\theta$, other entries zero, and choose $a_\theta > 0$ so $\|P_\theta\|_2 = 1$. Since $N_\theta^3 = 0$,

$$\exp(sP_\theta) = e^{-s/4}\left[I + sa_\theta N_\theta + \tfrac{1}{2}(sa_\theta)^2 N_\theta^2\right]. \tag{3}$$

All eigenvalues equal $-1/4$, but $\|\exp(P_\theta)\|_2$ increases from 1.225180 to 1.397337 over five shapes between 0 and $\pi/4$ (Fig. 5). Eigenvalues and the largest singular value therefore do not determine finite-time gain. Other singular values and geometry co-vary; this standard illustrative construction does not prove a generative benefit from any particular constraint.

7 Interpretation and limits

The evidence separates three questions: approximating a hidden linear flow, controlling a gated layer's sensitivity, and learning a useful generative velocity field. Improving one does not automatically improve the others. For example, Ring8 trained K4's exact substitution lowers oracle MSE from 0.03438 to 0.03371, yet worsens SW_2. Error location and propagation are plausible explanations, not isolated causes.

All training uses one seed; layers and checkpoints are correlated, and test sets have been repeatedly inspected. No statistical significance or independent-seed confidence interval is claimed. The toy networks have no attention and predict velocity, unlike JiT's clean-image prediction. There is no new image/FID evidence. These experiments support a mechanism-focused progress report, not a finished architectural contribution.

8 Hypotheses for the next two weeks

The next training direction remains to be selected. Two focused tests are proposed:

1. **Does input gating add more than a scalar duration?** Compare the original gate with fixed $s = 0.6$ and one learned scalar per layer, first at K2. Retain both FFN baselines and seed 0; fix a validation endpoint rule before training. A consistent input-gate advantage would support adaptive duration; matching performance would favor the simpler scale explanation.
2. **Is large P gain harmful mainly where the gate correction is large?** Before adding a constraint, stratify held-out native probes by gate-correction norm and measure local input-perturbation amplification through the full block. Test whether the gate term explains additional variation beyond $\|P\|_2$. If it does not, prioritize operator-only explanations. If it does, design one controlled gate or operator intervention rather than combining several tricks.

Supporting numeric comparison

Model / updates	Gaussian	Ring8	Moons
SiLU / 5,000	0.0868	0.1631	0.0448
SwiGLU / 5,000	0.0746	0.2080	0.0302
ODE K2 / 5,000	0.0674	0.1848	0.0271
SwiGLU / 6,572	0.0658	0.1508	0.0624

Table 1: Fixed-endpoint SW_2; seed 0. The last row matches the ODE $K = 2$ forward-linear-MAC budget. MACs exclude backward, nonlinear and optimizer costs; this is not a wall-clock match.

Additional visualizations and examples

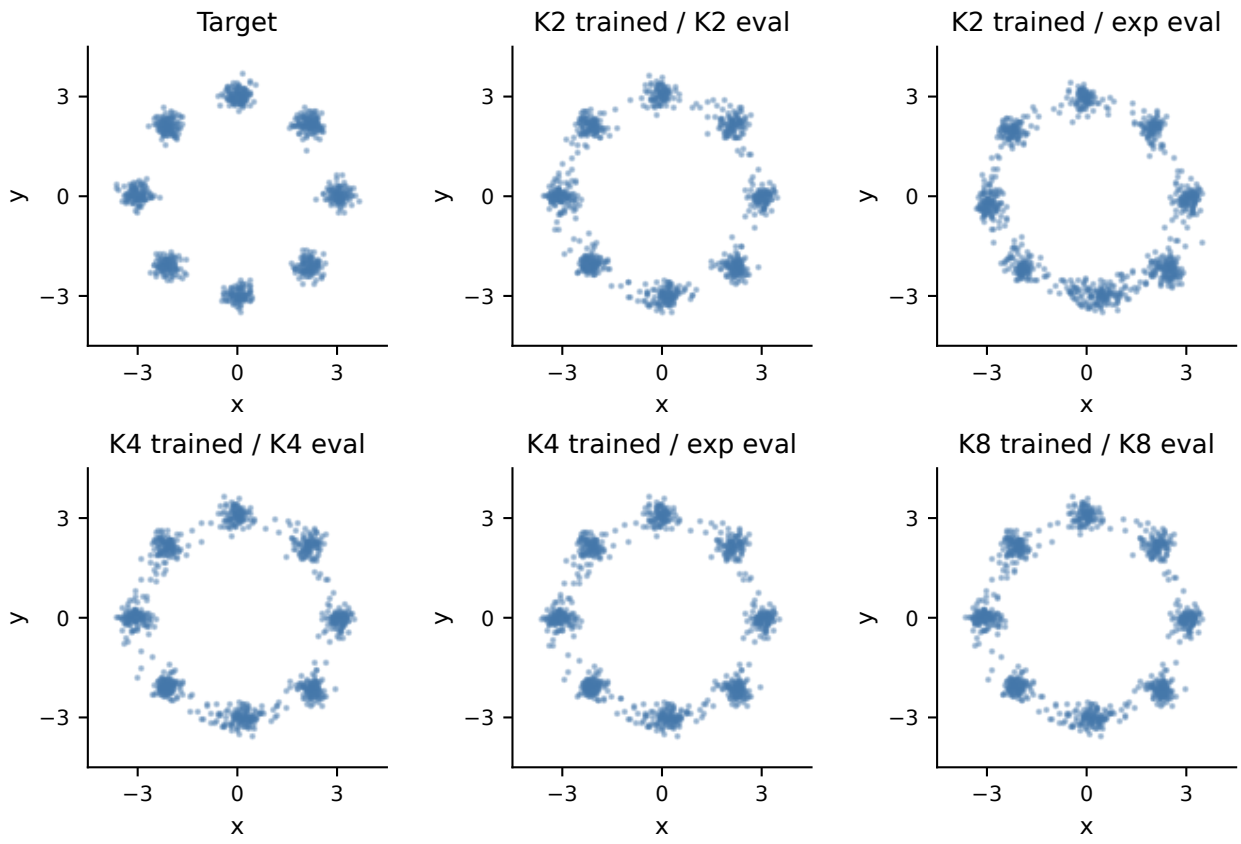


Figure 2: Ring8: first 1,000 target or generated points from fixed sample indices, with common axes. Model panels use the same source points, seed 0, update 5,000, and Heun32. The K2-to-exponential substitution alters the distribution; this is an inspection example, not a selected best/worst subset or a mode-coverage test.

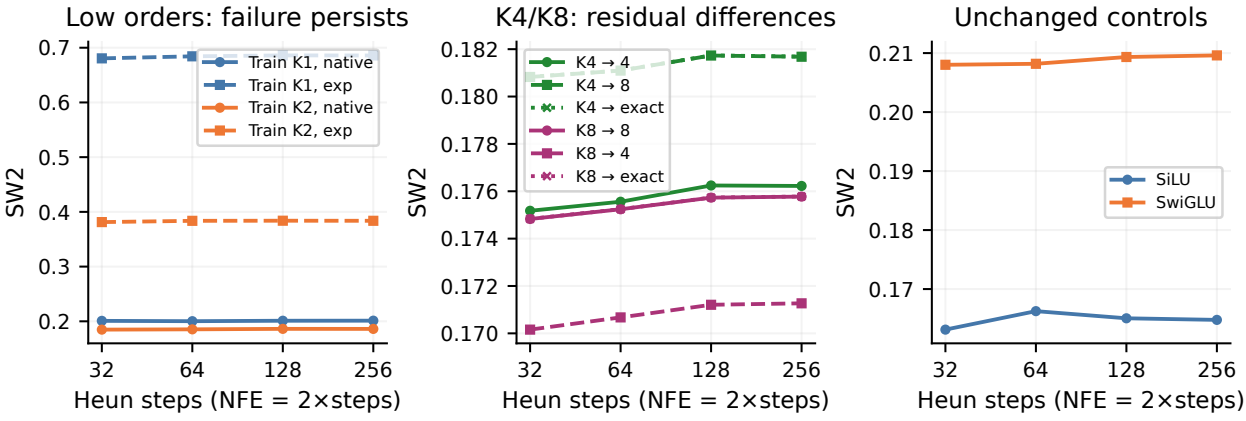


Figure 3: All twelve Ring8 model/order combinations at four Heun resolutions. Low-order exact-substitution failures and K4/K8 differences persist; baseline ordering is preserved. Fixed checkpoints and samples, no retraining. Horizontal axis is steps; the number of velocity evaluations is twice this value.

Additional diagnostics

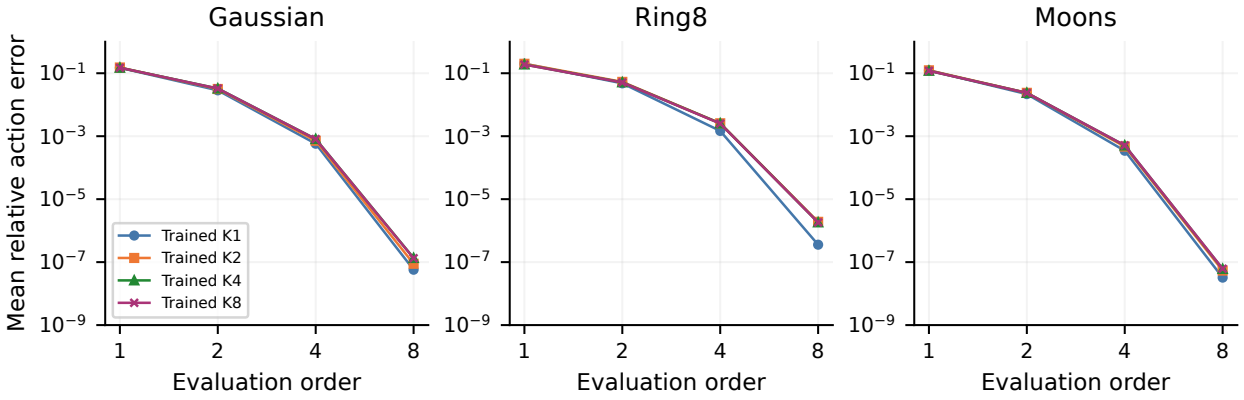


Figure 4: Local exponential-action discrepancy on identical captured native inputs and gates: means over 16 probes per layer and six layers, with a separate curve for each trained order. Higher evaluation order improves numerical accuracy; this does not imply better full-network generation. No independent-seed error bars are available.

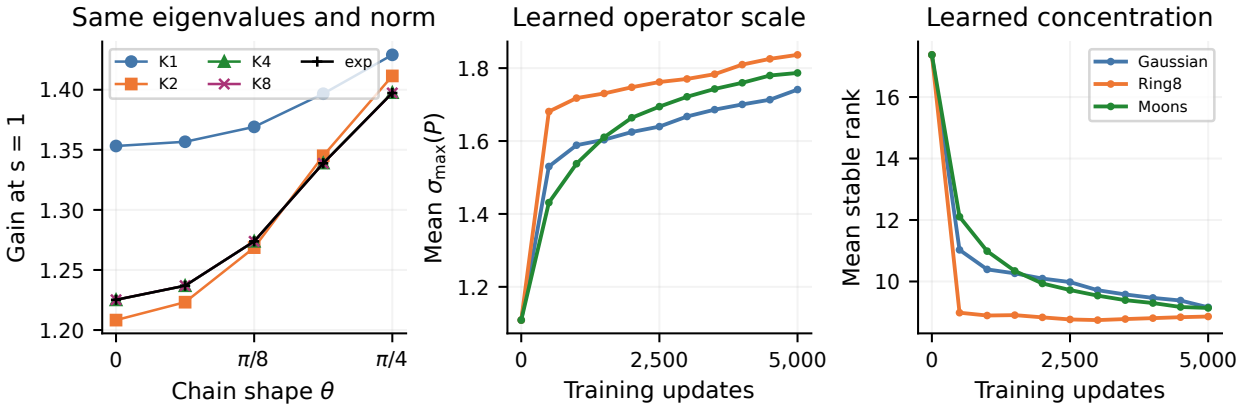


Figure 5: Left: the constructed family holds every eigenvalue at -0.25 and $\|P\|_2 = 1$; finite-time gains still differ. Right: K2 training trajectories show six-layer mean operator scale and stable rank for each target. Constructed matrices and learned networks are distinct analyses. Text ranges include all four trained orders.

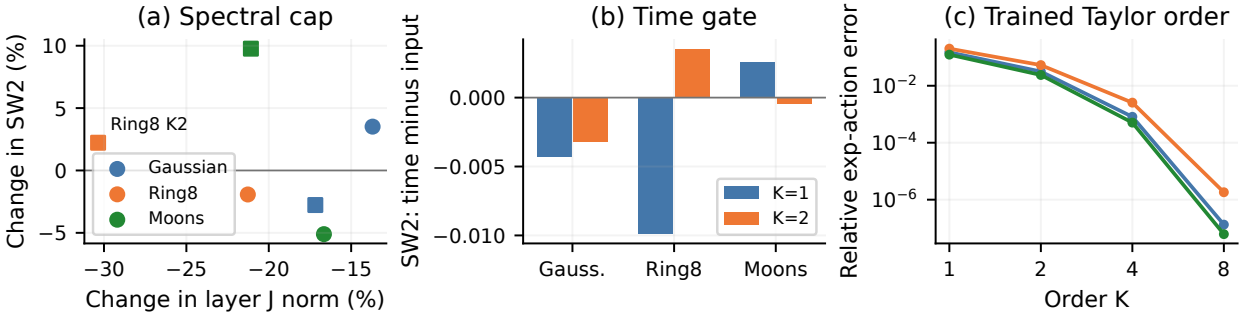


Figure 6: Earlier controls. (a) Free-to-capped changes; circles K1, squares K2. (b) Time-minus-input-gate SW_2; negative is better. (c) Action discrepancy for separately trained orders. Panels use seed 0 and 5,000 updates. These comparisons include retraining, unlike the frozen-order matrix in Fig. 1.